# Harsh Thomare

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## Education

**University of California, Irvine** Dec 2024
*B.S., Computer Science and Engineering, cum laude, GPA 3.8*

## Publications

**HELIOS: Hierarchical Graph Abstraction for Structure-Aware LLM Decompilation** Jan 2026
*LAST-X Workshop, NDSS Symposium 2026*
Yonatan Gizachew Achamyeleh, **Harsh Thomare**, Mohammad Abdullah Al Faruque. arXiv:2601.14598.

## Research Experience

**Research Collaborator** Apr 2026 - Present
*UCLA Security Lab (Prof. Yuan Tian)*

- Built RevBench, a containerized harness for evaluating LLM decompilers (LLM4Decompile, SK2Decompile) with taxonomy-mapped failure analysis
- Mapped decompiler failures to a 14-point defect taxonomy, showing single aggregate scores hide systematic weaknesses on structs, loops, and optimized code
- Building a weird-machine benchmark with contamination-controlled harnesses to test whether LLM agents reason about hidden program behavior

**Research Assistant** May 2025 - Nov 2025
*UC Irvine, AI Cyber-Physical Systems (AICPS) Lab (Prof. Mohammad Abdullah Al Faruque)*

- Co-developed HELIOS, a training-free LLM decompiler that turns Ghidra program structure into hierarchical context
- Evaluated HELIOS across six ISAs (-O0 to -O3); on x86-64, compiler feedback reached 94.9-96.5% compilability and 53.2-55.9% functional correctness

**Undergraduate Researcher** Sep 2022 - Mar 2023
*UC Irvine, AI Cyber-Physical Systems (AICPS) Lab (Prof. Mohammad Abdullah Al Faruque)*

- Built Raspberry Pi-based automation to collect differential pressure-sensor measurements for an acoustic side-channel study
- Evaluated speech-recovery models on recorded sensor data; acknowledged for contributions to "A Fly on the Wall," published at ACSAC 2024

## Work Experience

**Embedded Software Engineer** Oct 2026 - Present
*Mainspring Energy* *Menlo Park, CA*

- Developing motor control algorithms validated with SIL motor simulation testing
- Building bootloader and firmware update infrastructure for 1,000+ deployed devices using C/C++, Python, Yocto Linux, Docker, and STM32
- Building a data collection pipeline and leading data selection for an ML-based early-diagnosis tool for predictive maintenance

**Associate Embedded Software Engineer** Jul 2024 - Oct 2026
*Mainspring Energy* *Menlo Park, CA*

- Built firmware and OTA infrastructure for 1,000+ deployed devices using C/C++, Python, Yocto Linux, Docker, and STM32
- Developed real-time embedded systems and automated hardware testing, including multicore debugging and physics-based validation

**Electrical Engineering Intern** Jan 2024 - Apr 2024
*Second Order Effects*

- Built an FPGA data-capture pipeline for high-speed ADCs and verified it with SystemVerilog testbenches
- Wrote a Linux driver to expose captured samples for system-level testing

## Projects

**Qwen Mega-Kernel: Fused Triton Kernels**

- Fused Triton kernels for Qwen and improved achieved memory bandwidth by 30%

**IEEE Micromouse** Jan 2021 - Dec 2022

- Designed the PCB and built BLE logging for PID tuning

## Teaching

**Workshop Lead** Jan 2023 - Mar 2023
*IEEE Keebs Club*

- Founded the workshop program, designed the curriculum, and handled parts procurement
- Taught soldering and PCB design to 60 students through hands-on build workshops

## Honors and Awards

- Cum laude, B.S. Computer Science and Engineering, UC Irvine Dec 2024
- UROP Distinguished Research Fellowship, UC Irvine Jan 2023
- Sigma Prize Winner, Mental Health Innovation Challenge Jan 2022

## Service

**Ambassador** Jan 2016 - Present
*Akshaya Patra Foundation*

- Led a Bay Area sapling-sale initiative, coordinating volunteers and raising funds to feed 5,000 children

## Relevant Coursework

COMPSCI 190 - Efficient ML Computing (A)
COMPSCI 142A - Compilers and Interpreters (A)
COMPSCI 143A - Principles of Operating Systems (A)
EECS 112 - Organization of Digital Computers (A)